

VIRGINIA COMMONWEALTH UNIVERSITY

ASSIGNMENT REPORT

Course Code: MBAV 443M

Course Title: Statistical Modeling and AI-Driven Analytics

Assignment No.: Assignment 1

Student Name: R Pranav

Program: MBA Specialisation

Submission Date: 15/6/26

Assignment 1 MBAV 443M

1. Introduction

The Online Retail dataset contains transactional records from an online retail business. The dataset includes invoice information, stock codes, product descriptions, quantities purchased, invoice dates, unit prices, customer identifiers, and country information. The objective of this assignment is to perform exploratory data analysis (EDA) to understand the structure, quality, and characteristics of the dataset and identify meaningful patterns before further analytical modeling.

2. Data Understanding and Quality Assessment

The dataset contains 541,909 observations and 8 variables. Variables include both numerical and categorical attributes relevant to retail transactions.

Data quality assessment included verifying data types, analyzing missing values, and checking for duplicate records. The dataset contains 136,534 missing values in total, with 135,080 missing values in CustomerID. Duplicate record analysis identified 5,268 duplicate rows.

The dataset is generally suitable for analysis, although missing customer identifiers, duplicate records, and negative transaction quantities should be considered during preprocessing.

3. Methodology

The analysis was performed using Python with Pandas, NumPy, Matplotlib, and Seaborn. The analytical workflow consisted of:

1. Dataset loading and inspection
2. Data type verification
3. Missing value assessment
4. Duplicate record identification
5. Summary statistical analysis
6. Distribution analysis
7. Outlier detection
8. Relationship exploration
9. Data visualization

4. Visualization

A series of histograms, boxplots, scatterplots, and country-level transaction charts were generated. The visualizations revealed highly skewed distributions for Quantity and UnitPrice, substantial outliers, and a strong concentration of transactions originating from the United Kingdom.

5. Key Insight

The most significant insight obtained from the analysis is that the dataset is dominated by United Kingdom transactions, while transaction quantities and prices exhibit substantial skewness and extreme outliers. Negative quantities appear to represent returns or cancellations. Despite data quality issues, the dataset provides valuable information for customer analytics, sales forecasting, and business intelligence applications.

6. Conclusions

This exploratory analysis provided an understanding of the structure, quality, and characteristics of the Online Retail dataset. Summary statistics and visual inspection revealed meaningful business patterns and highlighted several data quality concerns, including missing values, duplicate records, and extreme observations.

The findings demonstrate the importance of exploratory data analysis as a preliminary step before predictive modeling. Future work may involve customer segmentation, demand forecasting, recommendation systems, and sales prediction models.